

Zero-Shot vs. Fine-Tuned: A Systematic Evaluation of Encoder, Decoder, and LLM Architectures for Clinical Text Classification

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Abstract

This study presents a systematic comparison of zero-shot large language models (LLMs) versus fine-tuned transformer architectures for single-label clinical text classification across three real-world, imbalanced healthcare datasets. We evaluate three architectural paradigms: encoder-based (PubMedBERT), encoder-decoder (T5-base), and decoder-only LLM (Mistral-7B with LoRA fine-tuning), against a zero-shot GPT-4 baseline using structured prompt engineering.

Our results demonstrate that domain-specific fine-tuned encoders (PubMedBERT) achieve superior efficiency-performance trade-offs (F1 0.63–0.97), while zero-shot LLMs provide rapid deployment capabilities without labeled data (F1 0.48–0.71). We introduce a reproducible zero-shot evaluation framework with optimized system prompts, constraint-based output formatting, and statistical significance testing. This work provides actionable guidance for model selection in resource-constrained clinical NLP deployments, balancing accuracy, latency, and computational cost.

Keywords: zero-shot classification, parameter-efficient fine-tuning, LoRA, clinical NLP, imbalanced classification, prompt engineering

I. Introduction & Background

1.1 The Zero-Shot vs. Fine-Tuning Paradigm

Recent advances in natural language processing have created a fundamental tension between two approaches: zero-shot LLM inference, which leverages pre-trained knowledge through prompt engineering without task-specific training data, and supervised fine-tuning (SFT), which adapts smaller models to specific domains using labeled examples .

Zero-shot models like GPT-4 and Mistral-7B generalize across tasks through instruction following, making them attractive for rapid prototyping in low-resource settings .

However, emerging evidence indicates that fine-tuned encoder models (e.g., BERT variants) consistently outperform zero-shot approaches when even modest task-specific data is available, particularly in specialized domains like healthcare .

Recent large-scale comparisons reveal that zero-shot LLMs perform competitively on sentiment analysis but frequently underperform on non-sentiment, multi-class, or domain-specific classification tasks requiring granular semantic distinctions .

Furthermore, synthetic LLM-generated training data rarely exceeds the performance of well-tuned encoder models on specialized tasks .

1.2 Research Objectives

This project addresses three critical gaps in the current literature:

1. Architectural Comparison: Systematic evaluation across encoder (PubMedBERT), encoder-decoder (T5), and decoder-only (Mistral-7B) paradigms under identical data conditions
2. Zero-Shot Methodology: Reproducible prompt engineering framework with constraint-based output formatting and statistical validation

- 3. Production Considerations: Analysis of inference latency, memory footprint, and cost-efficiency for real-world deployment scenarios

We focus on single-label clinical text classification—a high-stakes domain where model selection directly impacts downstream decision-making. Our evaluation spans three distinct clinical NLP tasks: medical literature categorization, drug review sentiment analysis, and mental health diagnostic screening.

II. Methodology

2.1 Datasets & Preprocessing

We utilized three publicly available clinical text datasets, each presenting unique challenges in class imbalance and linguistic complexity:

Dataset	Task Type	Classes	Imbalance Ratio	Preprocessing Strategy
Medical Abstracts [HuggingFace]	Multi-label → Single-label	5 (disease categories)	1:8	Random tie-breaking for multi-label instances; stratified sampling
Drug Reviews [Kaggle]	Ordinal regression → Classification	5 (1–5 star ratings)	1:12	Text cleaning (HTML tags, special chars); concatenation of <code>review drug_name condition</code>
Mental Health [HuggingFace]	Multi-class diagnostic	4 (DSM-5-TR categories)	1:15	DSM-5-TR consolidation; hybrid resampling (target n=16,040 per class)

Table 1

Class Imbalance Mitigation: For Mental Health, we implemented hybrid random resampling: majority classes were undersampled to the "Normal" class count (16,040),

while minority classes were oversampled with replacement to the same threshold. This preserved clinical validity while enabling model convergence.

2.2 Baseline: Logistic Regression with TF-IDF

We established a non-neural baseline using Logistic Regression with TF-IDF vectorization (max_features=10,000, n-grams=1–2). This provided a reproducible lower bound for performance comparison:

- Medical Abstracts: F1 = 0.54
- Drug Reviews: F1 = 0.55
- Mental Health: F1 = 0.93

The strong baseline on Mental Health (F1 0.93) reflects the dataset's clear lexical patterns, while lower scores on Medical Abstracts and Drug Reviews indicate greater semantic complexity requiring contextual modeling.

III. Zero-Shot Classification Framework

3.1 Prompt Engineering Strategy

We designed a structured zero-shot prompting framework based on current best practices for clinical NLP. Our approach incorporates three critical elements: (1) expert persona conditioning, (2) constrained output formatting, and (3) label definitions with decision boundaries.

System Prompt Template (GPT-4/Mistral-7B-Instruct)

Python

```
SYSTEM_PROMPT = """You are an expert clinical NLP classifier with domain
expertise in {domain}.
Your task is to analyze the provided clinical text and classify it into
exactly one of the following categories:

{label_definitions}
```

CLASSIFICATION RULES:

1. Select ONLY ONE label from the list above
2. Base your decision on explicit clinical evidence in the text
3. If multiple categories apply, choose the most clinically urgent or primary diagnosis
4. Respond with ONLY the label name, no explanation, no punctuation

OUTPUT FORMAT: <CATEGORY_NAME>

EXAMPLE OUTPUT: {example_label}

Text to classify:

{text}

Label:""

Domain-Specific Label Definitions

Medical Abstracts (5-class):

Python

```
LABEL_DEFS_MEDICAL = """
- NEOPLASMS: Cancers, tumors, oncological conditions, malignant growths
- DIGESTIVE: Gastrointestinal, liver, pancreatic, colorectal conditions
- NERVOUS: Neurological, psychiatric, neurodegenerative disorders
- CARDIOVASCULAR: Heart disease, vascular conditions, circulatory disorders
- GENERAL_PATHOLOGICAL: Infections, inflammation, metabolic disorders, other
"""
```

Drug Reviews (5-class):

Python

```
LABEL_DEFS_DRUGS = """
- VERY_NEGATIVE (1-star): Severe side effects, ineffective treatment, dangerous reactions
- NEGATIVE (2-star): Unpleasant side effects, limited efficacy, not recommended
- NEUTRAL (3-star): Mixed results, moderate efficacy, tolerable side effects
"""
```

```
- POSITIVE (4-star): Effective treatment, manageable side effects, would recommend
- VERY_POSITIVE (5-star): Highly effective, minimal side effects, life-changing benefits
"""
```

Mental Health (4-class):

Python

```
LABEL_DEFS_MENTAL = """
- DEPRESSIVE_SPECTRUM: Major depression, dysthymia, persistent depressive symptoms, hopelessness
- ANXIETY_STRESS: Generalized anxiety, panic disorder, PTSD, acute stress reactions
- BIPOLAR_PERSONALITY: Bipolar I/II, cyclothymia, borderline personality, mood instability
- NORMAL: No clinical symptoms, typical stress responses, healthy coping mechanisms
"""
```

3.2 Zero-Shot Inference Pipeline

Python

```
import openai
from typing import List, Dict

class ZeroShotClassifier:
    def __init__(self, model: str = "gpt-4", temperature: float = 0.0):
        self.model = model
        self.temperature = temperature # Deterministic for classification

    def classify(self, text: str, domain: str, labels: Dict) -> Dict:
        prompt = self._build_prompt(text, domain, labels)

        response = openai.ChatCompletion.create(
            model=self.model,
            messages=[
                {"role": "system", "content": "You are a precise clinical classifier."},
                {"role": "user", "content": prompt}
            ],
            temperature=self.temperature,
            max_tokens=10, # Constrain output length
```

```

        logprobs=True,      # Enable confidence estimation
        top_logprobs=5      # Capture top-k token probabilities
    )

    prediction = response.choices[0].message.content.strip()
    confidence =
self._extract_confidence(response.choices[0].logprobs)

    return {
        "prediction": prediction,
        "confidence": confidence,
        "prompt_tokens": response.usage.prompt_tokens,
        "completion_tokens": response.usage.completion_tokens
    }

def _build_prompt(self, text: str, domain: str, labels: Dict) -> str:
    return SYSTEM_PROMPT.format(
        domain=domain,
        label_definitions=labels["definitions"],
        example_label=labels["examples"][0],
        text=text[:2000] # Truncate long clinical notes
    )

```

3.3 Constraint-Based Output Parsing

To ensure valid predictions and enable quantitative comparison, we implemented strict output constraints:

1. Token Constraint: `max_tokens=10` prevents verbose explanations
2. Temperature=0.0: Ensures deterministic, reproducible outputs
3. Logit Bias: Applied +100 bias to newline token to prevent multi-line responses
4. Post-hoc Validation: Regex matching against valid label set with fuzzy string matching for robustness

Python

```

def validate_prediction(raw_output: str, valid_labels: List[str]) -> str:
    """Normalize and validate zero-shot model output."""
    cleaned = raw_output.strip().upper().replace(" ", "_")

    # Exact match
    if cleaned in valid_labels:
        return cleaned

```

```
# Fuzzy match using rapidfuzz for robustness
from rapidfuzz import process
match, score, _ = process.extractOne(cleaned, valid_labels)
return match if score > 80 else "INVALID"
```

3.4 Zero-Shot Results

Table 2

Dataset	Model	Accuracy	Macro F1	Weighted F1	Avg. Confidence	Cost/1K samples
Medical Abstracts	GPT-4	0.58	0.52	0.56	0.71	\$8.50
Medical Abstracts	Mistral-7B-ZS	0.52	0.48	0.51	0.64	\$0.12
Drug Reviews	GPT-4	0.61	0.59	0.60	0.68	\$6.20
Drug Reviews	Mistral-7B-ZS	0.54	0.51	0.53	0.61	\$0.10
Mental Health	GPT-4	0.71	0.70	0.71	0.82	\$4.80
Mental Health	Mistral-7B-ZS	0.64	0.63	0.64	0.75	\$0.08

Key Findings:

- Prompt sensitivity: Zero-shot performance varied $\pm 8\%$ across prompt variations; structured constraints improved consistency by 15% vs. free-form prompting
- Confidence calibration: GPT-4 showed better calibration (confidence correlated with accuracy) than Mistral-7B-Instruct
- Cost-efficiency: Mistral-7B-Instruct achieved 85% of GPT-4's performance at 1.4% of the cost

IV. Fine-Tuned Models

4.1 PubMedBERT: Domain-Specific Encoder

Architecture: `microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract`
(110M parameters)

Rationale: Pre-trained on 14M PubMed abstracts, providing inherent biomedical knowledge

Fine-Tuning Configuration:

Python

```
from transformers import AutoModelForSequenceClassification,
TrainingArguments

model = AutoModelForSequenceClassification.from_pretrained(
    "microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract",
    num_labels=num_classes,
    id2label=id2label,
    label2id=label2id
)

training_args = TrainingArguments(
    output_dir="./pubmedbert_results",
    learning_rate=2e-5,                # Conservative LR for stability
    per_device_train_batch_size=32,
    per_device_eval_batch_size=64,
    num_train_epochs=3,                # Early stopping on validation loss
    weight_decay=0.01,
    evaluation_strategy="epoch",
    save_strategy="epoch",
    load_best_model_at_end=True,
    metric_for_best_model="f1_macro",
    max_grad_norm=1.0,                # Gradient clipping
    warmup_ratio=0.1                  # Linear warmup
)
```

Optimization Strategy: We employed stratified k-fold cross-validation (k=5) to handle dataset shift between train/test splits, particularly critical for Medical Abstracts where pre-defined splits showed distribution mismatch.

4.2 T5-Base: Text-to-Text Transfer

Architecture: `t5-base` (220M parameters)

Approach: Text-to-text formulation framing classification as conditional generation:

`input: <text> target: <label>`

Hyperparameter Configuration:

- Max sequence length: 128 tokens (efficiency-accuracy trade-off)
- Learning rate: 5e-5 with linear decay
- Batch size: 16 (gradient accumulation steps=2 effective batch=32)
- Epochs: 3–5 with early stopping (patience=2)

Imbalance Handling: Unlike PubMedBERT, T5 showed minimal sensitivity to sampling strategies. We tested:

- Class-weighted loss: Weighted CrossEntropy (inverse frequency)
- Upsampling: Random duplication of minority classes
- Downsampling: Random removal from majority classes
- Baseline: No sampling (best performance)

Results indicated T5's text-to-text objective provides inherent regularization against class imbalance, likely due to the decoder's language modeling objective acting as an auxiliary task.

4.3 Mistral-7B: Parameter-Efficient Fine-Tuning

Architecture: `mistralai/Mistral-7B-v0.1` (7B parameters)

Method: LoRA (Low-Rank Adaptation) with 4-bit quantization

Configuration:

Python

```

from peft import LoraConfig, get_peft_model,
prepare_model_for_kbit_training
from transformers import BitsAndBytesConfig

# 4-bit quantization for memory efficiency
bnb_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_use_double_quant=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.bfloat16
)

# LoRA configuration
lora_config = LoraConfig(
    r=16, # Rank (higher = more capacity)
    lora_alpha=32, # Scaling factor (alpha/r = 2)
    target_modules=["q_proj", "k_proj", "v_proj", "o_proj"],
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM"
)

# Training configuration
training_args = TrainingArguments(
    per_device_train_batch_size=4, # Limited by A100 40GB VRAM
    gradient_accumulation_steps=4, # Effective batch = 16
    num_train_epochs=2, # LoRA converges faster
    learning_rate=2e-4, # Higher LR for adapters
    fp16=True,
    optim="paged_adamw_8bit", # Memory-efficient optimizer
    group_by_length=True # Dynamic padding
)

```

Computational Profile:

- Hardware: NVIDIA A100 40GB
- Training time: 2.5–4.5 hours per dataset (2 epochs)
- Memory usage: ~28GB VRAM (4-bit base + LoRA adapters)
- Checkpoint size: 134MB (LoRA weights only vs. 14GB full model)

V. Results & Analysis

5.1 Performance Comparison

Table 3

Model	Medical Abstracts F1	Drug Reviews F1	Mental Health F1	Avg. Inference Time	Model Size
Logistic Regression	0.54	0.55	0.93	2ms	45MB
Zero-Shot GPT-4	0.56	0.60	0.71	850ms	N/A (API)
Zero-Shot Mistral-7B	0.51	0.53	0.64	120ms	14GB
T5-Base (Fine-tuned)	0.64	0.49	0.97	15ms	850MB
PubMedBERT (Fine-tuned)	0.63	0.66	0.97	8ms	440MB
Mistral-7B + LoRA	0.48	0.70	0.92	95ms	134MB*

*LoRA adapter size only; base model shared across tasks

5.2 Architectural Analysis

PubMedBERT demonstrated the most consistent performance across domains, leading on Medical Abstracts (F1 0.63) and Mental Health (F1 0.97). Its biomedical pre-training provided clear advantages for domain-specific terminology. However, we observed mild overfitting on Medical Abstracts (validation F1 0.73 vs. test F1 0.63), attributable to distribution shift between the provided train/test splits.

T5-Base excelled on structured text with clear lexical patterns (Medical Abstracts F1 0.64, Mental Health F1 0.97) but struggled with subjective, noisy data (Drug Reviews F1 0.49). The text-to-text framework's rigidity limited performance on ambiguous user-generated content where sentiment boundaries are fuzzy.

Mistral-7B + LoRA achieved the best performance on Drug Reviews (F1 0.70), capturing nuanced sentiment and side-effect descriptions. Its strong contextual understanding compensated for data sparsity, though at significant computational cost. Performance on Medical Abstracts was moderate (F1 0.48), likely due to the domain shift from general pre-training to specialized biomedical literature.

5.3 Statistical Significance

We employed paired bootstrap resampling (10,000 iterations) to validate performance differences :

- PubMedBERT vs. Zero-Shot GPT-4: Significant improvement on Medical Abstracts ($p < 0.001$, 95% CI: [0.04, 0.11])
- Mistral-7B + LoRA vs. PubMedBERT: Significant improvement on Drug Reviews ($p = 0.003$, 95% CI: [0.02, 0.08])
- T5 vs. PubMedBERT: No significant difference on Mental Health ($p = 0.42$)

5.4 Production Deployment Considerations

Table 4

Metric	PubMedBERT	T5-Base	Mistral-7B + LoRA	GPT-4 API
Latency (p99)	12ms	22ms	145ms	1200ms
Throughput	850 req/s	420 req/s	85 req/s	20 req/s
Memory/Instance	2GB	3GB	16GB	N/A
Cost/1M predictions	\$0.80	\$1.20	\$4.50	\$8,500
Offline Capability	✓	✓	✓	✗

Recommendation Matrix:

- High-volume, latency-sensitive: PubMedBERT (best efficiency-performance trade-off)
- Subjective/ambiguous content: Mistral-7B + LoRA (superior semantic understanding)
- Rapid prototyping/no training data: Zero-shot GPT-4 (immediate deployment, highest cost)
- Resource-constrained environments: T5-Base (balanced capabilities, moderate footprint)

VI. Discussion

6.1 When to Fine-Tune vs. Zero-Shot

Our results align with recent findings that zero-shot LLM classification, while operationally convenient, typically underperforms fine-tuned models on specialized tasks . The performance gap is most pronounced in:

- Domain-specific terminology: PubMedBERT's biomedical pre-training provided 12% F1 improvement over zero-shot on Medical Abstracts
- Class imbalance: Fine-tuned models with resampling outperformed zero-shot by 18% on the imbalanced Drug Reviews dataset
- Structured outputs: Constrained classification with mutually exclusive labels favors encoder architectures

However, zero-shot approaches remain valuable for:

- Cold-start scenarios: No labeled data available for fine-tuning
- Dynamic label spaces: Labels change frequently (e.g., emerging drug categories)
- Prototyping: Rapid validation of task feasibility before investing in annotation

6.2 Prompt Engineering Impact

Our structured prompting framework improved zero-shot consistency by 15% compared to basic instruction prompts . Key design decisions:

1. Expert persona conditioning: Improved clinical reasoning accuracy by 8%
2. Constrained output formatting: Reduced invalid predictions from 12% to <1%

3. Label definitions with decision boundaries: Critical for fine-grained distinctions (e.g., anxiety vs. stress)

Notably, requesting chain-of-thought reasoning degraded performance by 3%—contrary to findings in mathematical reasoning, classification tasks benefit from direct prediction without intermediate steps .

6.3 Parameter-Efficient Fine-Tuning Efficacy

LoRA adaptation achieved 94% of full fine-tuning performance (where tested) with only 1.9% of trainable parameters. This enables:

- Multi-tenant deployments: Single base model with task-specific adapters
- Edge deployment: 134MB adapters vs. 14GB full model
- Rapid iteration: 2-hour training cycles vs. 8+ hours for full fine-tuning

VII. Conclusion

This study provides empirical guidance for architectural selection in clinical text classification. Our findings support a hierarchical decision framework:

1. Data availability > 1,000 samples: Fine-tune domain-specific encoder (PubMedBERT)
2. Ambiguous/subjective content: Deploy LoRA-adapted LLM (Mistral-7B)
3. No training data available: Use structured zero-shot with GPT-4 for prototyping, Mistral-7B-Instruct for cost-sensitive production
4. Latency-critical applications: Avoid API-based LLMs; prefer distilled encoders

The "best" model is context-dependent: PubMedBERT offers optimal efficiency-performance balance for structured clinical text, while Mistral-7B + LoRA provides superior semantic understanding for nuanced patient-generated content. Zero-shot LLMs bridge the gap when annotation is infeasible, albeit with measurable accuracy trade-offs.

VIII. Future Work

1. Advanced Imbalance Techniques: Implement focal loss and label smoothing to improve minority class recall
2. Synthetic Data Augmentation: Evaluate LLM-generated training data quality for low-resource classes
3. Multi-Task Learning: Joint training across all three datasets to improve generalization
4. Attention Visualization: Implement LIME/SHAP for clinical decision interpretability
5. Quantization-Aware Training: Deploy INT8 quantized models for edge clinical devices

Appendix I: Complete Zero-Shot Prompts

A. Medical Abstracts Classification

Python

```
MEDICAL_SYSTEM_PROMPT = """You are a biomedical research classifier.
Analyze the medical abstract and classify it into exactly one disease
category.

CATEGORIES:
- NEOPLASMS: Cancers, tumors, oncological conditions, malignant neoplasms
- DIGESTIVE: Gastrointestinal, hepatic, pancreatic, colorectal conditions
- NERVOUS: Neurological, psychiatric, neurodegenerative, CNS disorders
- CARDIOVASCULAR: Cardiac, vascular, hypertension, circulatory conditions
- GENERAL_PATHOLOGICAL: Infectious, inflammatory, metabolic, endocrine,
other

RULES:
1. Select exactly one category
2. Base decision on the primary disease focus of the research
3. If multiple conditions, choose the most severe or primary study
endpoint
4. Output ONLY the category name in UPPERCASE with underscores

EXAMPLE OUTPUT: NEOPLASMS

Abstract:
{text}
```



```
Category:""
```

B. Drug Review Sentiment

Python

```
DRUG_SYSTEM_PROMPT = """You are a pharmacovigilance analyst. Classify this drug review into a sentiment category based on efficacy and side effects.
```

```
RATING CATEGORIES:
```

- VERY_NEGATIVE: Severe adverse effects, treatment failure, discontinuation due to safety
- NEGATIVE: Significant side effects, limited therapeutic benefit, would not recommend
- NEUTRAL: Mixed results, moderate efficacy, tolerable but not optimal
- POSITIVE: Effective treatment, manageable side effects, satisfactory outcome
- VERY_POSITIVE: Excellent efficacy, minimal/no side effects, highly recommend

```
CLASSIFICATION CRITERIA:
```

- Consider both effectiveness and tolerability
- Weight long-term outcomes over initial reactions
- Flag reports of serious adverse events as VERY_NEGATIVE regardless of efficacy

```
OUTPUT FORMAT: Return ONLY the category name.
```

```
Review:
```

```
{text}
```

```
Sentiment:""
```

C. Mental Health Screening

Python

```
MENTAL_HEALTH_PROMPT = """You are a clinical psychologist conducting preliminary screening. Classify this text into one diagnostic category.
```

```
DSM-5-TR CATEGORIES:
```

- DEPRESSIVE_SPECTRUM: Major depression, persistent depressive disorder, hopelessness, anhedonia, suicidal ideation

- ANXIETY_STRESS: Generalized anxiety, panic attacks, PTSD, acute stress, excessive worry, hypervigilance
- BIPOLAR_PERSONALITY: Bipolar I/II, cyclothymia, borderline personality, emotional dysregulation, mania
- NORMAL: Typical stress responses, situational sadness, healthy coping, no clinical symptoms

DECISION GUIDELINES:

- Focus on duration (>2 weeks for depression), severity (functional impairment), and symptom clusters
- Distinguish between clinical disorders and normal emotional responses
- When symptoms overlap, prioritize the primary presenting complaint

OUTPUT: Provide ONLY the category name.

Patient text:
{text}

Classification:""

Appendix II: Implementation Details

Reproducibility Checklist

- Random seeds: 42 (NumPy, PyTorch, transformers)
- Hardware: NVIDIA A100 40GB (training), V100 16GB (inference)
- Software: Python 3.9, PyTorch 2.0.1, Transformers 4.35.0, PEFT 0.6.0
- Data splits: Stratified 80/10/10 (train/val/test) with fixed random state

Ethical Considerations

- All datasets are publicly available with appropriate licenses
- Mental health classifications are for research purposes only; not intended for diagnostic use
- No protected health information (PHI) was used in this study

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Author Contributions

[Kirthi Shanbhag]: Designed zero-shot evaluation framework, implemented prompt engineering strategies, conducted statistical analysis, and led paper writing. Developed reproducible inference pipelines for GPT-4 and Mistral-7B.

[Helen Lu]: Led PubMedBERT fine-tuning and Medical Abstracts dataset analysis. Implemented baseline models and stratified sampling strategies. Contributed to results interpretation and presentation materials.

[Monica Martin]: Led Mistral-7B LoRA fine-tuning and Mental Health dataset processing. Implemented DSM-5-TR consolidation and hybrid resampling. Conducted computational efficiency profiling.
